亞洲大學
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Midterm Project Report

Advanced Computer Programming

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Chapter 1 Introduction

1.1 Github

1) Personal Github Account: https://github.com/khanyasarpak

2) Group Project Repository: The report files and source code are prepared for upload to the group repository inside the midterm folder as required by the assignment.

1.2 Overview

This midterm assignment uses Python and the Scrapy framework to scrape repository information from my GitHub page. The program applies CSS selectors, XPath, and regular expressions to extract repository URL, About, Last Updated, Languages, and Number of Commits, and then exports the collected data to an XML file. For repositories without a description, the program uses the repository name as the About value when the repository is not empty. For empty repositories, the program stores Languages and Number of Commits as None.

Chapter 2 Implementation

2.1 Spider Class

The program is implemented in a Scrapy spider class named GithubReposSpider. The spider receives the GitHub username as an argument, builds the repositories page URL, and starts crawling from https://github.com/khanyasarpak?tab=repositories.

2.1.1 Key Components

• Spider name: github_repos

• Allowed domain: github.com

• Custom settings: download delay, user agent, and robots setting

2.2 parse() Method

The parse() method reads the repositories list page and selects each repository card. It extracts the repository link and the last updated value from the card, then follows the repository link to the detail page for further extraction.

2.3 parse_repo() Method

The parse_repo() method extracts the detailed fields from every repository page. It reads the repository URL, repository name, About section, programming language information, and number of commits. The method also checks whether the repository is empty. If the About section is blank and the repository is not empty, the repository name is used as the fallback About value. If the repository is empty, Languages and Number of Commits are stored as None.

2.4 Export Command

The spider is executed from the project root using the following command:

py -m scrapy crawl github_repos -a username=khanyasarpak -O repos.xml

Chapter 3 Results

3.1 Extracted Repository Information

The spider executed successfully and generated an XML output file named repos.xml. In the current run, one repository record was extracted from the GitHub account. The About field used the repository name as fallback because the repository page did not provide a description. The number of commits was extracted as 6, and the language field in the current output appeared as None.

Field	Extracted Value
URL	https://github.com/khanyasarpak/first_project
About	first_project
Last Updated	2026-04-15T11:18:47Z
Languages	None
Number of Commits	6

3.2 Execution Screenshots

The following screenshots show the successful execution of the spider, the generated XML file, the main spider implementation file, and the GitHub repository page used for validation.

```
C:\Users\AK\github_midterm>py -m scrapy crawl github_repos -a username=khanyasarpak -O repos.xml
2026-04-16 14:42:44 [scrapy.utils.log] INFO: Scrapy 2.15.0 started (bot: github_midterm)
2026-04-16 14:42:44 [scrapy.utils.log] INFO: Versions:
{'lxml': '6.0.4',
 'libxml2': '2.11.9',
 'cssselect': '1.4.0',
 'parsel': '1.11.0',
 'w3lib': '2.4.1',
 'Twisted': '25.5.0',
 'Python': '3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 '
           '64 bit (AMD64)]',
 'pyOpenSSL': '26.0.0 (OpenSSL 3.5.6 7 Apr 2026)',
 'cryptography': '46.0.7',
 'Platform': 'Windows-10-10.0.19045-SP0'}
2026-04-16 14:42:44 [scrapy.crawler] DEBUG: Using AsyncCrawlerProcess
2026-04-16 14:42:44 [asyncio] DEBUG: Using selector: SelectSelector
2026-04-16 14:42:44 [scrapy.addons] INFO: Enabled addons:
[]
2026-04-16 14:42:44 [scrapy.utils.log] DEBUG: Using reactor: twisted.internet.asyncioreactor.AsyncioSelectorReactor
2026-04-16 14:42:44 [scrapy.utils.log] DEBUG: Using asyncio event loop: asyncio.windows_events._WindowsSelectorEventLoop
2026-04-16 14:42:44 [scrapy.extensions.telnet] INFO: Telnet Password: f7682909c6a2150d
2026-04-16 14:42:44 [scrapy.middleware] INFO: Enabled extensions:
['scrapy.extensions.corestats.CoreStats',
 'scrapy.extensions.logcount.LogCount',
 'scrapy.extensions.telnet.TelnetConsole',
 'scrapy.extensions.feedexport.FeedExporter',
 'scrapy.extensions.logstats.LogStats']
2026-04-16 14:42:44 [scrapy.crawler] INFO: Overridden settings:
{'BOT_NAME': 'github_midterm',
 'CONCURRENT_REQUESTS_PER_DOMAIN': 1,
 'DOWNLOAD_DELAY': 1,
 'FEED_EXPORT_ENCODING': 'utf-8',
 'NEWSPIDER_MODULE': 'github_midterm.spiders',
 'SPIDER_MODULES': ['github_midterm.spiders'],
 'USER_AGENT': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 '
               '(KHTML, like Gecko) Chrome/124.0.0.0 Safari/537.36'}
2026-04-16 14:42:45 [scrapy.middleware] INFO: Enabled downloader middlewares:
['scrapy.downloadermiddlewares.offsite.OffsiteMiddleware',
```

Figure 1. Terminal output showing successful spider execution and XML export.

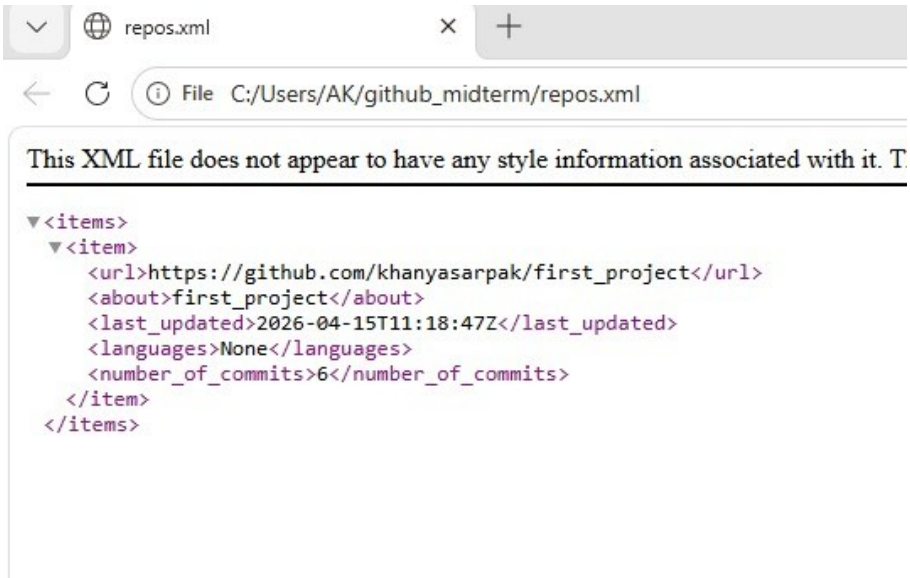

Figure 2. Generated repos.xml file opened in the browser.

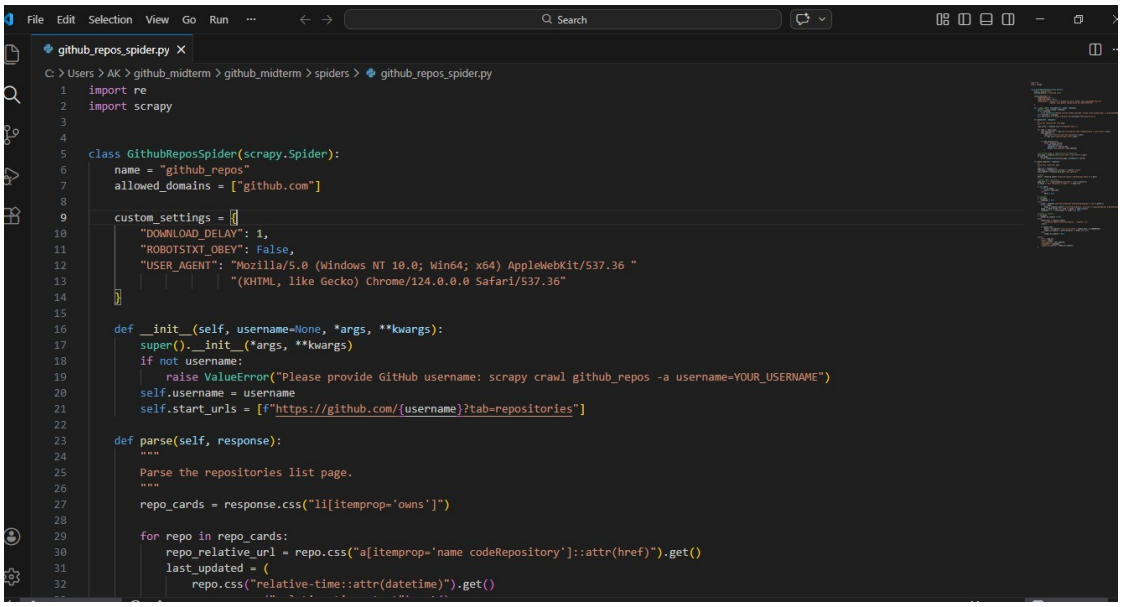

Figure 3. Main spider implementation file in the project folder.

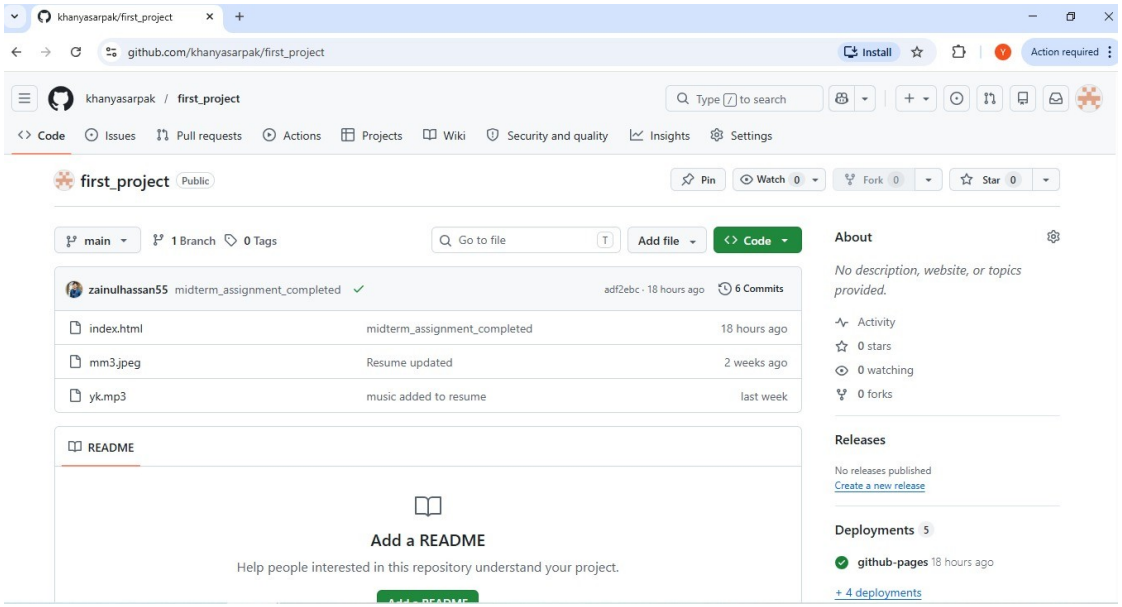

Figure 4. GitHub repository page used to validate description and commit count.

Chapter 4 Conclusions

This assignment was completed successfully by creating a Scrapy spider that scraped repository information from GitHub and exported the results to an XML file. The final program was able to extract the repository URL, About value, last updated timestamp, and number of commits from the tested repository. The report, XML file, and source code archive are ready for submission to TronClass and to the group GitHub repository.

Files prepared for submission:

• PDF report

• Source code archive (.zip or .rar)

• Generated XML file (repos.xml)